INTEGRAL CLOSURE COMMUTES WITH SMOOTH PULLBACK FOR ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6121-6129 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 6131-6138, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Integral closure of a quasi-coherent sheaf of algebras, smooth morphisms of algebraic spaces, and compatibility with etale localisation; the reduction is retained.

Source reference. Original lemma algebra-lemma-integral-closure-commutes-smooth: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6138–6138. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-definition-integral-closure: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6138–6138. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a smooth morphism of algebraic spaces over S. Let $\mathcal{A}$ be a quasi-coherent sheaf of $\mathcal{O}_X$-algebras. The integral closure of $\mathcal{O}_Y$ in $f^*\mathcal{A}$ is equal to $f^*\mathcal{A}'$ where $\mathcal{A}' \subset \mathcal{A}$ is the integral closure of $\mathcal{O}_X$ in $\mathcal{A}$.*

Proof. By our construction of the integral closure, see Morphisms of Spaces, Definition spaces-morphisms-definition-integral-closure, this reduces immediately to the case where X and Y are affine. In this case the result is Algebra, Lemma algebra-lemma-integral-closure-commutes-smooth. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.